TIØ4145 Corporate Finance Chapters 21,22

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Option Valuation

- Chapter 21
  - 21.1 The Binomial Option Pricing Model
  - **21.2 The Black-Scholes Option Pricing Model**
  - **21.3 Risk-Neutral Probabilities**
  - **21.4 Risk and Return of an Option**
  - ~~21.5 Beta of Risky Debt~~

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Learning objective

- Use the **Black-Scholes** Option Pricing **formula** to calculate the value of a **call option on a non-dividend-paying stock**.
- Use the **Black-Scholes** Option Pricing **formula** to calculate the price of **a European put option on a non-dividend paying stock**.
- Compute the value of a **European option on a dividend-paying stock**.
- Define **the Black-Scholes replicating portfolio** for a call option on a non-dividend paying stock or a European put option on a non-dividend paying stock.
- Discuss what is meant by **risk-neutral probabilities**, and **show how these probabilities can be used** to price any other asset for which the payoffs in each state are known.
- **Define and calculate the risk-neutral probability** that the stock price will increase in a binomial tree.
- Calculate and interpret **the beta of an option**.

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The Black-Scholes Option Pricing Model

- Black-Scholes Option Pricing Model
  - A technique for pricing European-style options when the stock can be traded continuously.
  - It can be derived from the Binomial Option Pricing Model by allowing the length of each period to shrink to zero and letting the number of periods grow infinitely large.

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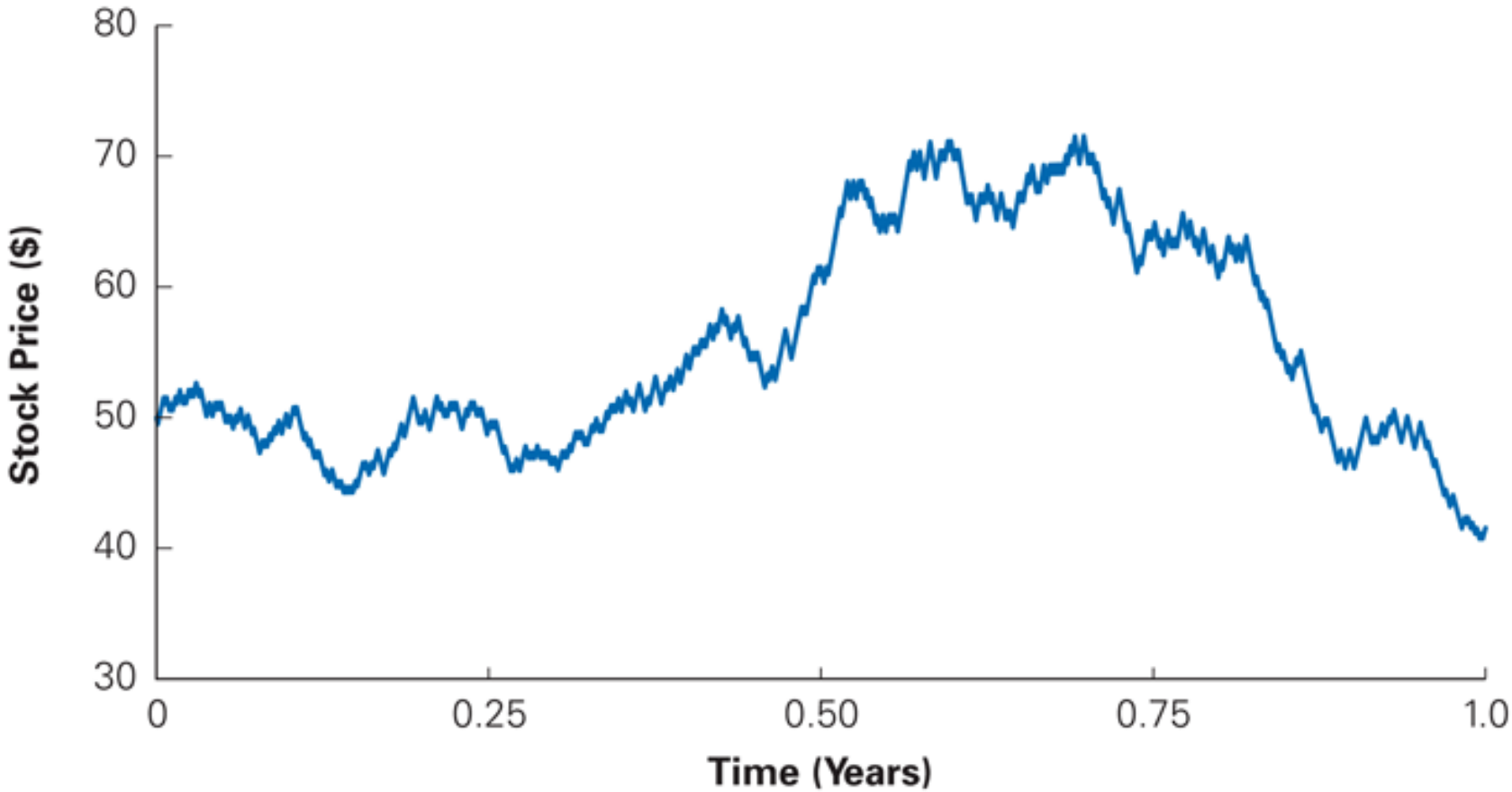
A Binomial Stock Price Path



The Black-Scholes Formula

- Black-Scholes Price of a Call Option on a Non-Dividend-Paying Stock:

$$C = S \times N(d_1) - PV(K) \times N(d_2) \qquad (21.7)$$

  - Where S is the current price of the stock, K is the exercise price, and N(d) is the cumulative normal distribution
  - Cumulative Normal Distribution is the probability that an outcome from a standard normal distribution will be below a certain value

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Normal Distribution

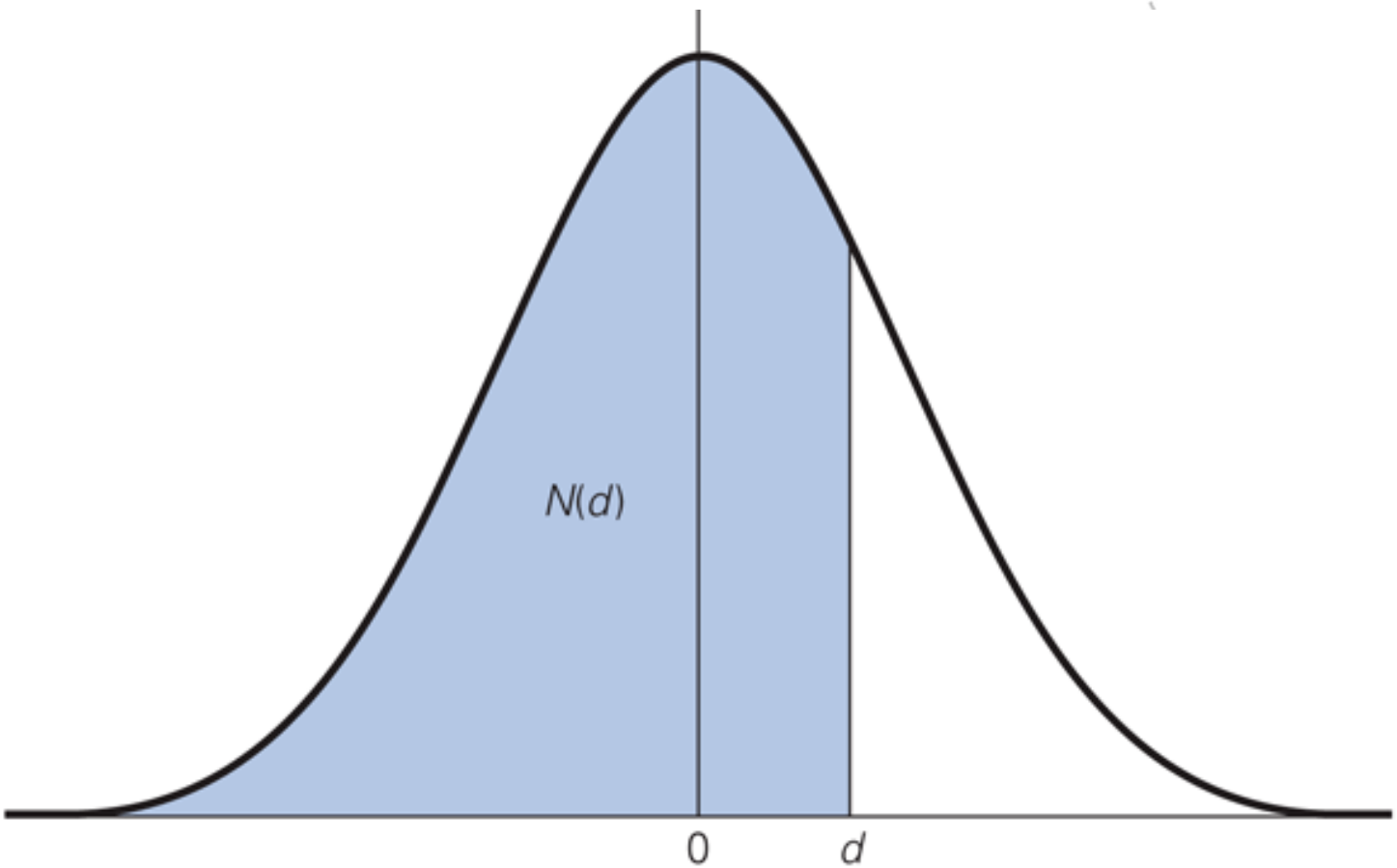

Figure 21.3

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The Black-Scholes Formula

$$d_1 = \frac{\ln[S / PV(K)]}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2} \quad and \quad d_2 = d_1 - \sigma\sqrt{T} \qquad (21.8)$$

- Where σ is the annual volatility, and T is the number of years left to expiration

- Note: Only five inputs are needed for the Black-Scholes formula.
  - Stock price
  - Strike price
  - Exercise date
  - Risk-free rate
  - Volatility of the stock

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Example

- Assume:
  - CLW Inc. does not pay dividends.
  - The standard deviation of CLW is 45% per year.
  - The risk-free rate is 5%.
  - CLW stock has a current price of $24.
- Using the Black-Scholes formula, what is the price for a ½ year American call option on CLW with a strike price of $30?

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Table 1: Table of the Standard Normal Cumulative Distribution Function $\Phi(z)$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483

Example

$$PV(K) = \frac{\$30}{1.05^{1/2}} = \$29.277$$

$$d_1 = \frac{\ln[\$24 / \$29.277]}{.45\sqrt{.5}} + \frac{.45\sqrt{.5}}{2} = -0.4658$$

$$d_2 = -0.4658 - .45\sqrt{.5} = -0.7840$$

$$N(-0.4658) = 0.3207 \qquad N(-0.7480) = 0.2165$$

$$C = \$24 \times 0.3207 - \$29.277 \times 0.2165 = \$1.36$$

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European Put Options

- Using put-call parity, the value of a European put option is $P = C - S + PV(K)$
- Black-Scholes Price of a European Put Option on a Non-Dividend-Paying Stock

$$P = PV(K)[1 - N(d_2)] - S[1 - N(d_1)] \qquad (21.9)$$

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The Black-Scholes Option Pricing Model

- The Black-Scholes option pricing model is derived assuming that the call is a European option
- The Black-Scholes option pricing model can be used to price
  - European call options on non-dividend paying stocks
  - American call options on non-dividend paying stocks
    - Price of American on non-dividend paying stock is equal to the otherwise equivalent European call option
  - European Put options on non-dividend paying stocks

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Dividend-Paying Stocks

- If *PV(Div)* is the present value of any dividends paid prior to the expiration of the option, then:

$$S^x = S - PV(Div) \qquad (21.10)$$

  - Where S^x is the price of the stock excluding any dividends
  - Because a European call option is the right to buy the stock without these dividends, it can be evaluated by using the Black-Scholes formula with S^x in place of S.

- A special case is when the stock will pay a dividend that is proportional to its stock price at the time the dividend is paid.
  - If q is the stock's (compounded) dividend yield until the expiration date, then:

$$S^x = S / (1 + q) \qquad (21.11)$$

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Implied Volatility

- Of the five required inputs in the Black-Scholes formula, only σ is not observable directly.
- Practitioners use two strategies to estimate the value of σ.
  - Use historical data
  - "Back out" the implied volatility
- Implied Volatility - the volatility of an asset's returns that is consistent with the quoted price of an option on the asset

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The Replicating Portfolio

- Given:
  - Option Price in the Binomial Model

$$C = S\Delta + B$$

- Then:
  - Black-Scholes Replicating Portfolio of a Call Option

$$\Delta = N(d_1)$$
$$B = -PV(K)N(d_2) \qquad (21.12)$$

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The Replicating Portfolio

- Option Delta (Δ)
  - The change in the price of an option given a $1 change in the price of the stock. The number of shares in the replicating portfolio for the option.
  - Note: Because Δ is always less than 1, the change in the call price is always less than the change in the stock price.

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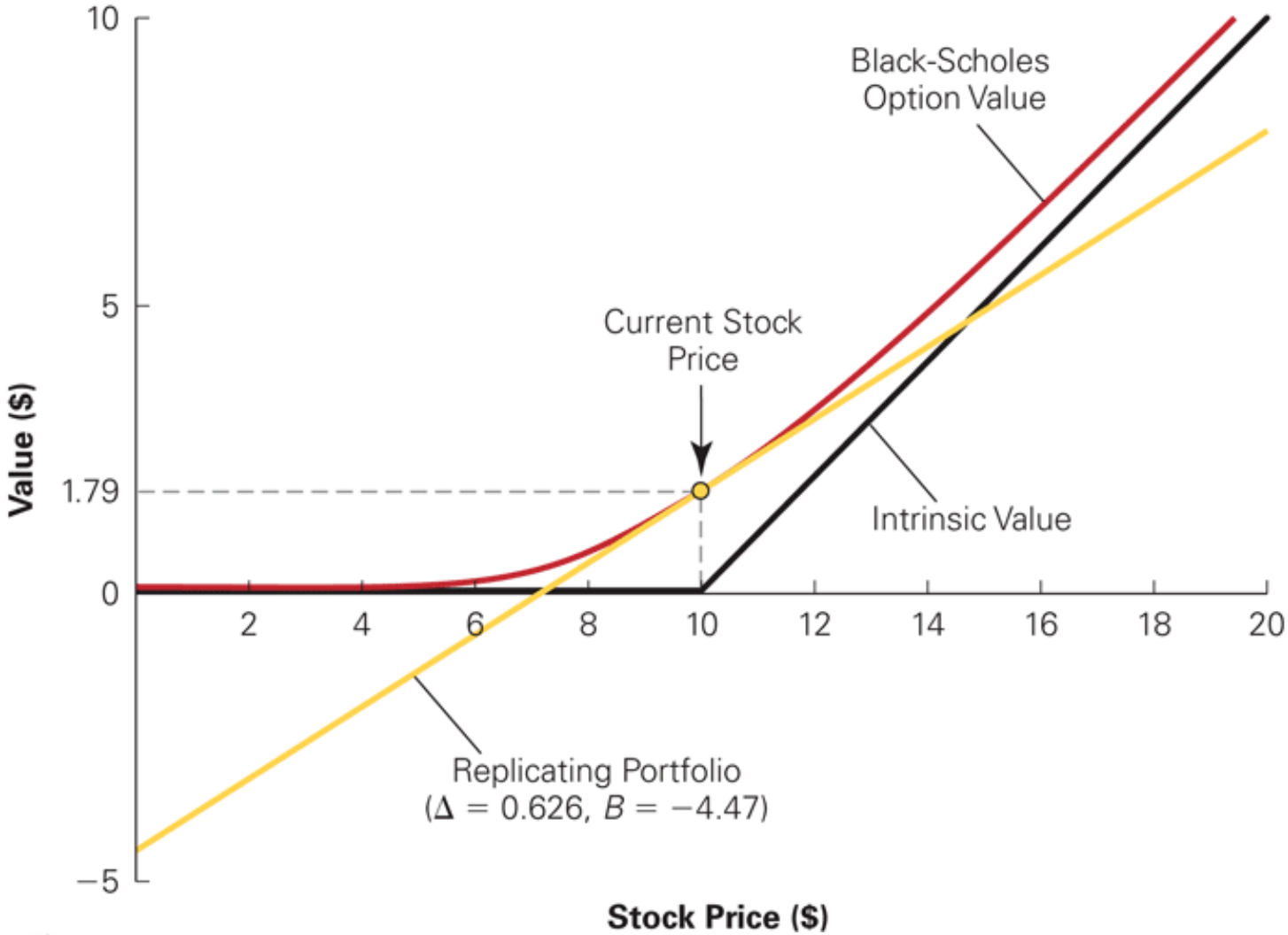
Replicating Portfolio for the Call Option



The Replicating Portfolio

- Note: The replicating portfolio of a call option always consists of a long position in the stock and a short position in the bond.
  - The replicating portfolio is a leveraged position in the stock.
  - A leveraged position in a stock is riskier than the stock itself, which implies that call options on a positive beta stock are more risky than the underlying stock and therefore have higher returns and higher betas.

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The Replicating Portfolio

- The replicating portfolio of a put option is calculated as
  - Black-Scholes Replicating Portfolio of a Put Option

$$\begin{aligned} \Delta &= -[1 - N(d_1)] \\ B &= PV(K)[1 - N(d_2)] \end{aligned} \qquad (21.13)$$

  - The replicating portfolio of a put option always consists of a long position in the bond and a short position in the stock,
  - implying that put options on a positive beta stock will have a negative beta.

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Risk-Neutral Probabilities

- If all market participants are risk neutral,
  - then all financial assets (including options) have the same cost of capital, the risk-free rate of interest.
- Assume a world consisting of only risk-neutral investors, and consider the original two-state example:
  - The stock price today is equal to $50.
  - In one period it will either go up by $10 or go down by $10.
  - The one-period risk-free rate of interest is 6%.

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Risk-Neutral Two-State Model

- If ρ is the probability that the stock price will increase, then $(1 - \rho)$ is the probability that it will go down.
  - The value of the stock today must equal the present value of the expected price next period discounted at the risk-free rate.

$$50 = \frac{60\rho + 40(1 - \rho)}{1.06} \qquad (21.14)$$

  - Solving for ρ yields $\rho = .65$
- The present value of the expected payouts is:

$$\frac{10(0.65) + 0(1 - 0.65)}{1.06} = 6.13 \qquad (21.15)$$

Risk-Neutral Two-State Model

- This is precisely the value calculated using the Binomial Option Pricing Model where it was not assumed that investors were risk neutral.
- Because no assumption on the risk preferences of investors is necessary to calculate the option price using either the Binomial Model or the Black-Scholes formula, the models work for any set of preferences, including risk-neutral investors.

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Implications of the Risk-Neutral World

- The Binomial and Black-Scholes models give the same option price no matter what the actual risk preferences and expected stock returns are.
- In the real world, investors are risk averse and require a positive risk premium to compensate for risk, while in a hypothetical risk-neutral world, investors do not.
- In other words, ρ is not the actual probability of the stock price increasing.
- It represents how the actual probability would have to be adjusted to keep the stock price the same in a risk-neutral world.

Implications of the Risk-Neutral World

- Risk-Neutral Probabilities
  - The probability of future states that are consistent with current prices of securities assuming all investors are risk neutral.
  - Probabilities ρ and $(1 - \rho)$ are risk-neutral probabilities.
  - Also known as State-Contingent Prices, State Prices, or Martingale prices

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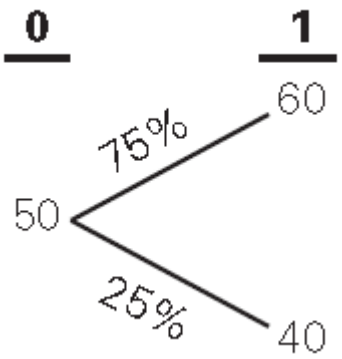
Implications of the Risk-Neutral World

- Assume from the previous example that the current price of $50 has a true probability of 75% of increasing to $60 and a true probability of 25% of decreasing to $40.



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Implications of the Risk-Neutral World

- To ensure that all assets in the risk-neutral world have an expected return equal to the risk-free rate:
  - relative to the true probabilities, the risk-neutral probabilities overweight the bad states and underweight the good states.

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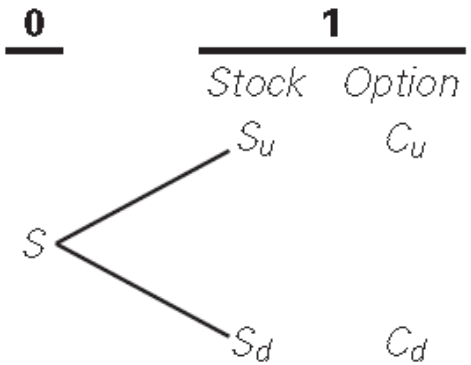
Risk-Neutral Probabilities and Option Pricing

- Consider again the general binomial stock price tree.



- Compute the risk-neutral probability that makes the stock's expected return equal to the risk-free interest rate:

$$\frac{\rho S_u + (1 - \rho) S_d}{S} - 1 = r_f$$

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Risk-Neutral Probabilities and Option Pricing

- Solving for the risk-neutral probability ρ yields:

$$\rho = \frac{(1 + r_f)S - S_d}{S_u - S_d} \qquad (21.16)$$

  - The value of the option can be calculated by computing its expected payoff using the risk-neutral probabilities, and discount the expected payoff at the risk-free interest rate.

- Derivative Security
  - A security whose cash flows depend solely on the prices of other marketed assets

- The probabilities in the risk-neutral world can be used to price any derivative security.

Risk-Neutral Probabilities and Option Pricing

- Monte Carlo Simulation:
  - A common technique for pricing derivative assets in which the expected payoff of the derivative security is estimated by calculating its average payoff after simulating many random paths for the underlying stock
  - In the randomization, the risk-neutral probabilities are used and so the average payoff can be discounted at the risk-free rate to estimate the derivative security's value.

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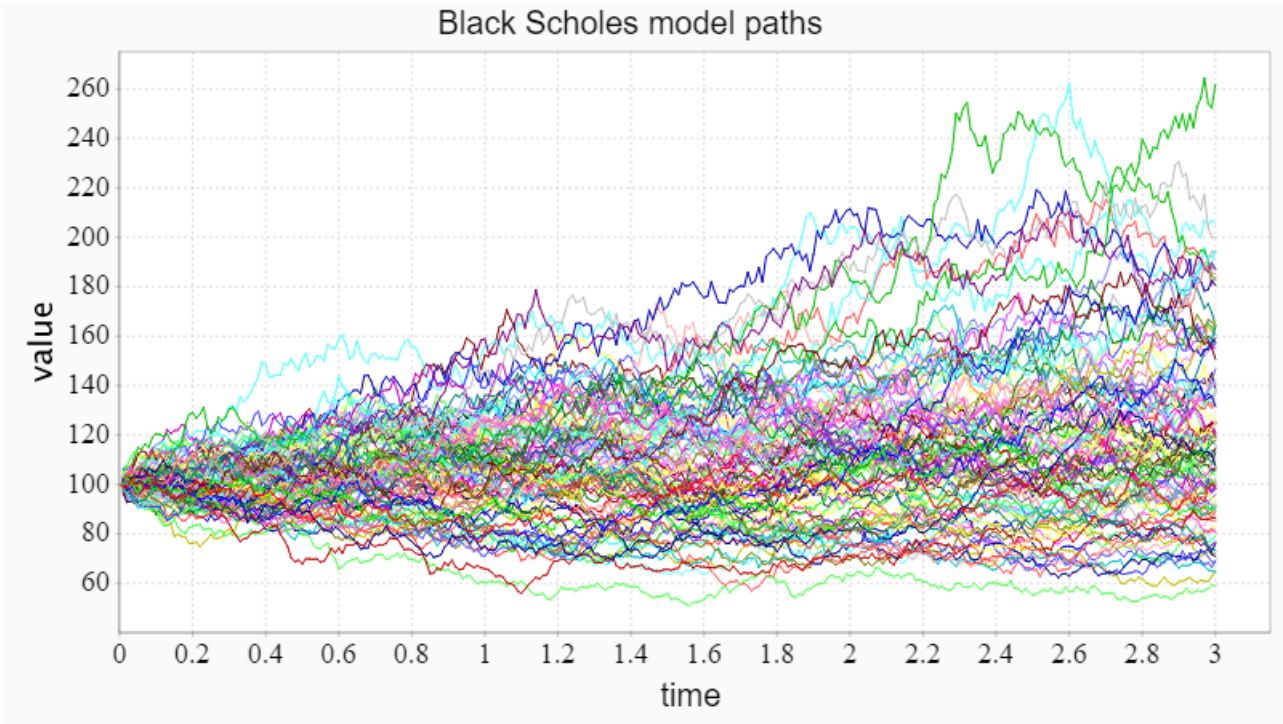
Monte Carlo Simulations



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Risk and Return of an Option

- To measure the risk of an option, we must compute the option beta.
  - This can be accomplished by computing the beta of the replicating portfolio.

$$\beta_{option} = \frac{S\Delta}{S\Delta + B}\beta_S + \frac{B}{S\Delta + B}\beta_B$$

  - Where β_S is the stock's beta and β_B is the bond's beta. Since the bond is riskless, $\beta_B = 0$:
- Beta of an Option

$$\beta_{option} = \frac{S\Delta}{S\Delta + B}\beta_S \qquad (21.17)$$

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Summary

- The Black-Scholes option pricing formula for the price of a call option on a non dividend-paying stock

$$C = S * N(d_1) - PV(K) * N(d_2) \qquad (21.7)$$

- where $N(d)$ is the cumulative normal distribution and

$$d_1 = \frac{\ln[S / PV(K)]}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2} \quad and \quad d_2 = d_1 - \sigma\sqrt{T} \qquad (21.8)$$

- The Black-Scholes formula for the price of a European put option on a non dividend-paying stock is

$$P = PV(K)[1 - N(d_2)] - S[1 - N(d_1)] \qquad (21.9)$$

- We can evaluate a European option on a stock that pays dividends using the Black-Scholes formula with Sx in place of S where $Sx = S - PV(Div)$ or $Sx = S/(1 + q)$ if the stock pays a (compounded) dividend yield of q prior to the expiration date
- The Black-Scholes replicating portfolio is
  - For a call option on a non-dividend-paying stock $\Delta = N(d1)$ and $B = -PV(K)N(d_2)$ (21.12)
  - For a European put option on a non-dividend-paying stock $\Delta = -[1 - N(d_1)]$ and $B = PV(K)[1 - N(d_2)]$ (21.13)
- Risk-neutral probabilities are the probabilities under which the expected return of all securities equals the risk-free rate. These probabilities can be used to price any other asset for which the payoffs in each state are known.
- In a binomial tree, the risk-neutral probability that the stock price will increase is given by

$$\rho = \frac{(1 + r_f)S - S_d}{S_u - S_d} \qquad (21.16)$$

- The beta of an option is the beta of the underlying stock times the option's leverage ratio:

$$\beta_{option} = \frac{S\Delta}{S\Delta + B}\beta_S \qquad (21.17)$$

Let’s do a QUIZ

- Log into Kahoot.it

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Real Options

- Chapter 22
  - 22.1 Real Versus Financial Options
  - 22.2 Decision Tree Analysis
  - 22.3 The Option to Delay an Investment Opportunity
  - 22.4 Growth and Abandonment Options
  - ~~22.5 Investments with Different Lives~~
  - ~~22.6 Optimally Staging Investments~~
  - ~~22.6 Rules of Thumb~~
  - 22.7 Key Insights from Real Options

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Learning objectives

- Define the term "**real option**".
- **Draw decision trees** to represent alternative decisions and potential outcomes in an uncertain economy.
- Describe three types of real options—**timing, growth, and abandonment**—and explain why it is important to consider those options when evaluating projects.
- Illustrate how, **given the option to wait, an investment** that currently has a negative NPV **can have a positive value**.
- Describe **situations in which the option** to wait **is most valuable**.

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Real Versus Financial Options

- Capital Budgeting
  - Has focused on the initial investment decision without explicitly considering future decisions that may be required over the life of a project.
  - Assumed that forecast of project's expected future cash flows already incorporated effect of future decisions made.
- Real Option
  - The right to make a particular business decision, such as a capital investment.
  - Take closer look how these cash flows are determined when a firm must react to changing business conditions.
  - Real options allow decision maker to choose the most attractive alternative after new information becomes available.

Real Options

- A real option is the right, but not the obligation, to take an action on a project or an asset (expand, defer, contract, abandon etc.) at some cost (exercise price)
- ROA applies financial theory to the valuation of non-financial assets
  - whose performance is affected by uncertainty
  - which require at least partly irreversible sunk costs
  - which offer a room for flexibility to change the course of the project over time

Uncertainty revelation → Ability to optimize actions → Additional value

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Decision Tree Analysis

- Assume Megan is financing part of her studies by running a small business. She purchases goods on eBay and resells them at swap meets.
  - Swap meets typically charge her $500 in advance to set up her small market stand.
  - Ignoring the cost of the market stand, if she goes to every meet, her average profit on the goods that she sells is $1100 per meet.

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Decision Tree Analysis

- Because the NPV of setting up a booth is $600, the optimal decision (shown in blue) would be to set up the booth.
  - $1100 – $500 = $600

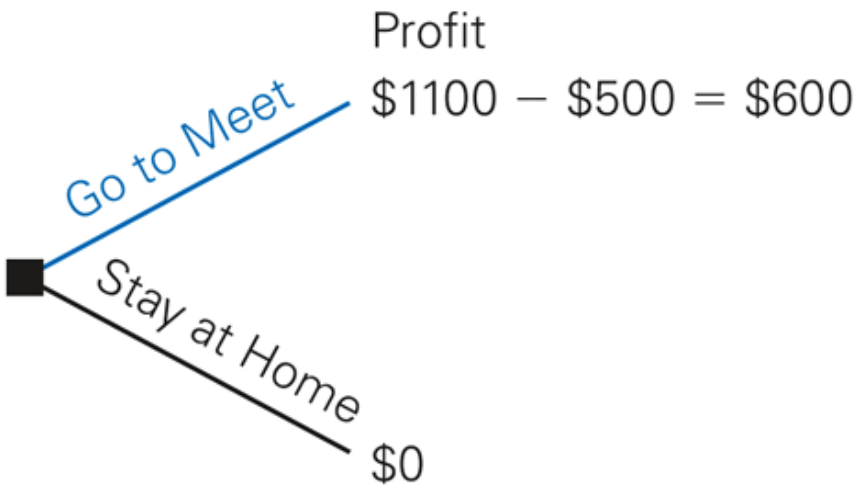

Figure 22.1

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Mapping Uncertainties on a Decision Tree

- Megan is aware that attendance at swap meets is weather dependent.
  - In good weather her profits are $1500, in bad weather, she will incur a loss of $100.
  - There is a 25% chance of bad weather.
- This adds another element of uncertainty for Megan to consider.

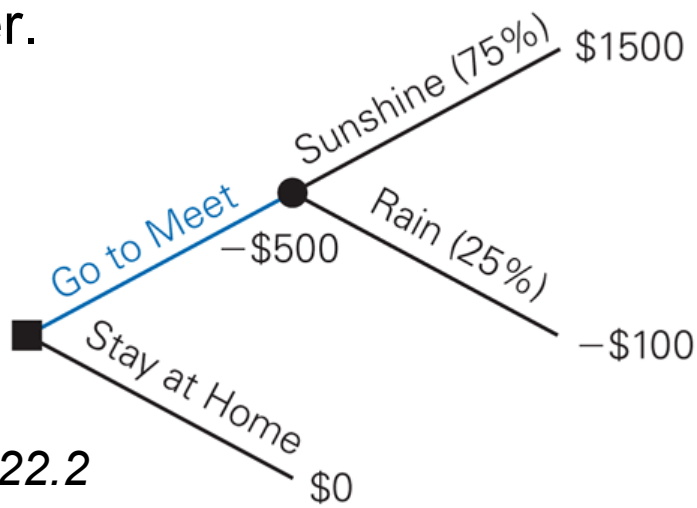

Figure 22.2

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Mapping Uncertainties on a Decision Tree

- Decision Nodes
  - A node on a decision tree at which a decision is made
  - Corresponds to a real option
- Information Nodes
  - A type of node on a decision tree indicating uncertainty that is out of the control of the decision maker
- In Megan's case:
  - The square node represents the decision to pay the fee and go to the swap meet or do nothing.
  - The round node represents the uncertain state of nature
    - In this case, Megan must commit to going to the meet before she knows what the weather will be.

Mapping Uncertainties on a Decision Tree

- In reality, Megan does not have to commit to going to the swap meet before she knows the weather conditions.
  - Megan understands that the $500 loss for the booth is unavoidable, but in bad weather she can simply stay home and not incur the additional $100 loss at the meet.

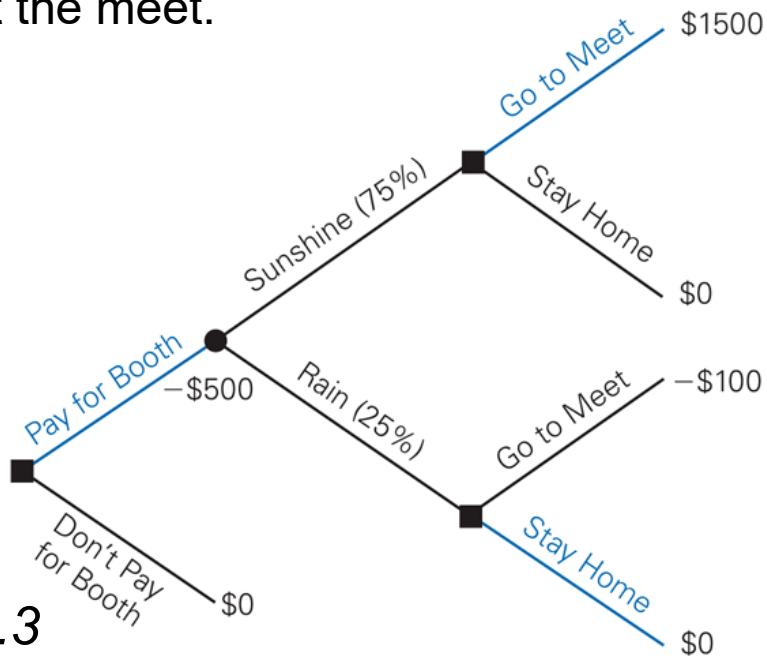

Figure 22.3

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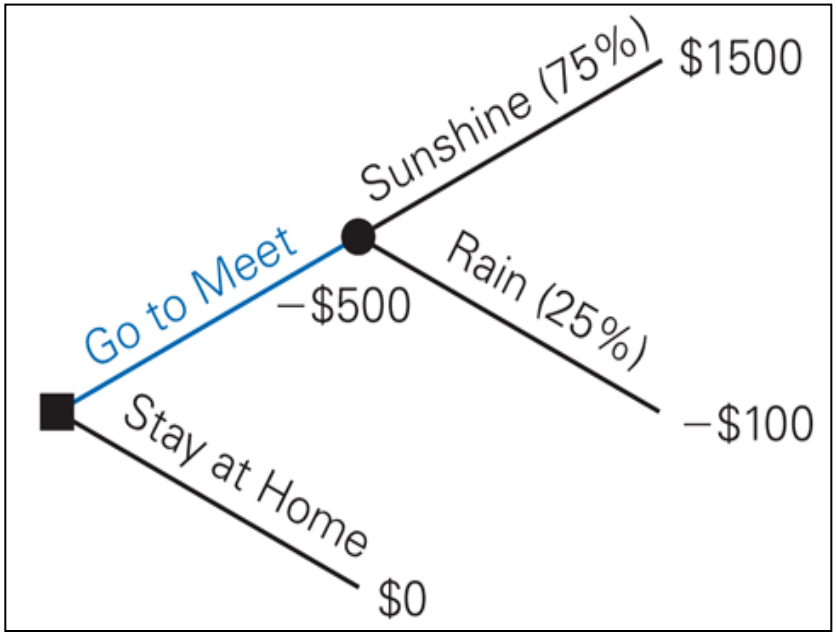
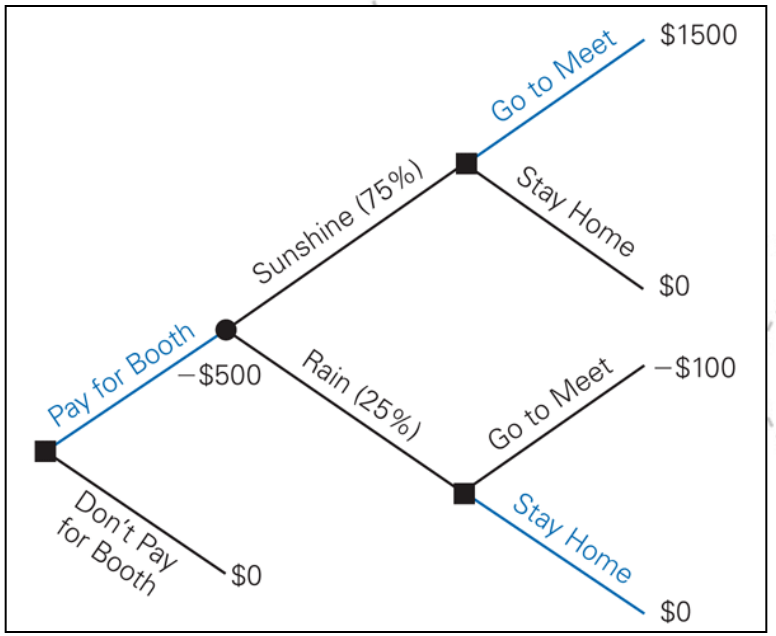
Real Options



Difference



- Megan's option to wait until she finds out what the weather is like before she decides whether she should go to the meet is a real option.
  - This flexibility has value to Megan.

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Real Options

- Assume Megan is risk neutral about the weather risk.
  - The value of the real option can be computed by comparing her expected profit without the real option to wait until the weather is revealed to the value with the option to wait.
- If Megan commits to go regardless of the weather, her expected profit is $1100.
  - 0.75 × $1500 + 0.25 × (–$100) = $1100
- However, if she goes only when the weather is good, her expected profit is $1125.
  - 0.75 × $1500 + 0.25 × $0 = $1125
  - **The value of the real option is the difference, $25.**

Real Options

- Many corporate investment decisions contain real options.
- Most frequently encountered kinds of real options in practice:
  - The option to delay an investment opportunity
  - The option to grow
  - The option to abandon an investment opportunity

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The Option to Delay an Investment Opportunity

- Megan's case: once the market stand is paid for, there is no cost to waiting to find out about the weather.
- In the real world, there is often a cost to delaying an investment decision.
- By choosing to wait for more information the firm gives up any profits the project might generate in the interim or/and gives competitors time to develop a competing product.
- The decision to wait involves tradeoff:
- Costs of waiting ↔ benefit of remaining flexible

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Investment as a Call Option

- Assume you are part of the management team of the energy company ANEO. Currently you consider the possibility to invest in a new wind park.
  - You agree that you will make the decision to invest either immediately or in exactly one year.
  - After 2016, you lose the possibility to invest in the wind park since shareholders are just willing to fund projects that are finalized before 2021 (due to the deadline of the elcertificate subsidies in Norway).
- Given the flexibility you have to delay investment for one year, what should you be willing to pay?
- When should you invest?

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Wind Park Investment Opportunity

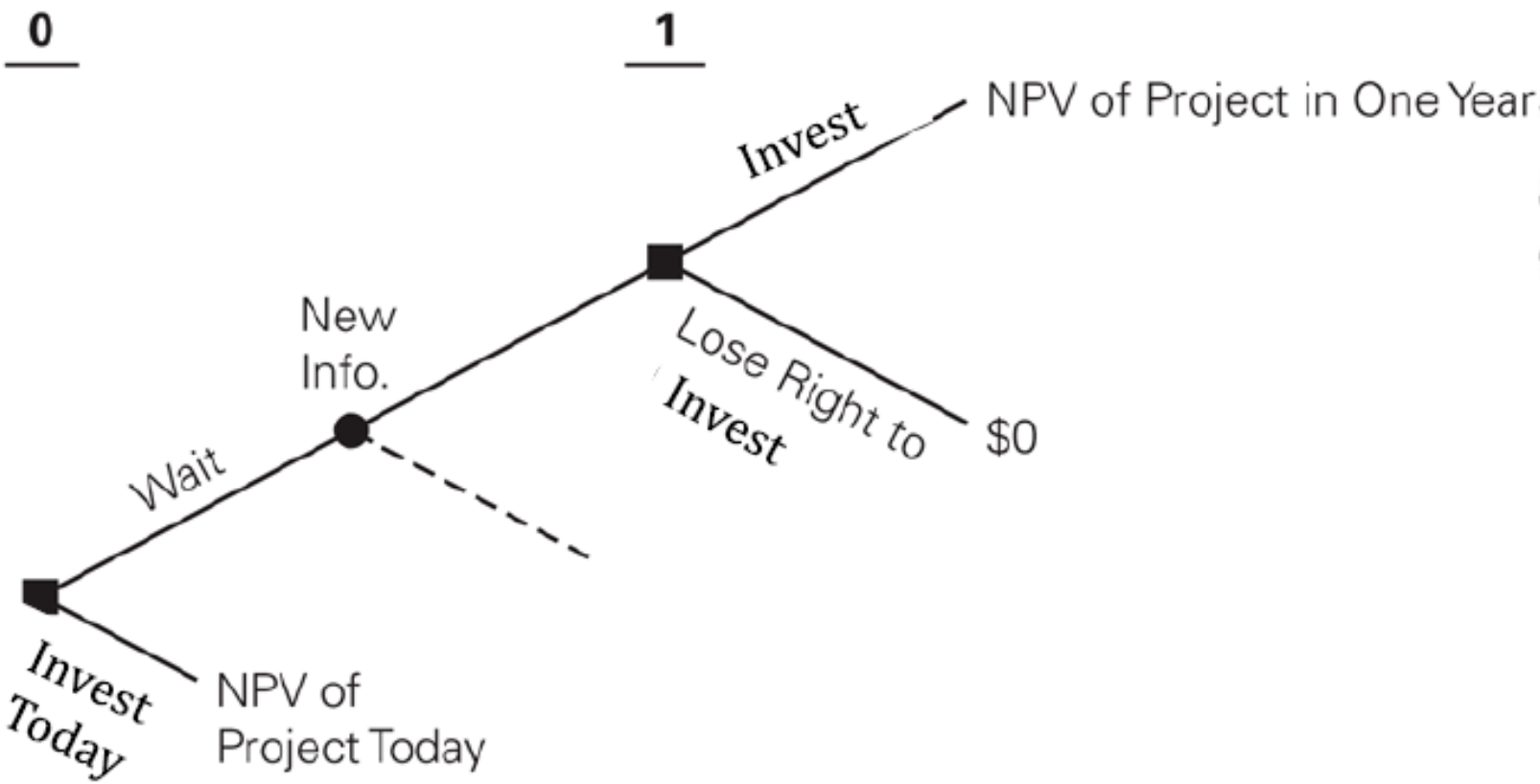

Figure 22.4

Investment as a Call Option

- How much you should pay for this opportunity?
  - It will cost 600 MNOK to build the wind park, whether you start building /invest now or in one year.
  - If you invest in the wind park immediately, you expect it to generate 50 MNOK in free cash flow the first year.
  - Future cash flows are expected to grow at a rate of 2% per year.
  - The cost of capital for this investment is 10%.

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Investment as a Call Option

- If the wind park would be operating today, its value would be:

$$V = \frac{50\ MNOK}{10\% - 2\%} = 625\ MNOK$$

- This would give an NPV of 25 MNOK.
  - 625 MNOK – 600 MNOK = 25 MNOK

- Given the flexibility you have to delay investment for one year, what should you be willing to pay?

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Investment as a Call Option

- The payoff if you delay is equivalent to the payoff of a one-year European call option on the wind park investment with a strike price of 600 MNOK.
- Assume
  - The risk-free interest rate is 4%.
  - The volatility is 30%.
  - If you wait to invest in the wind park you have an opportunity cost of 50 MNOK (the free cash flow in the first year).
  - In terms of a financial option, the free cash flow is equivalent to a dividend paid by a stock. The holder of a call option

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Black-Scholes Option Value Parameters for Evaluating a Real Option to Invest

Financial Option		Real Option	Example
Stock Price	S	Current Market Value of Asset	625 MNOK
Strike Price	K	Upfront Investment Required	600 MNOK
Expiration Date	T	Final Decision Date	1 year
Risk-Free Rate	r_f	Risk-Free Rate	4%
Volatility of Stock	σ	Volatility of Asset Value	30%
Dividend	Div	FCF Lost from Delay	50 MNOK

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Investment as a Call Option

- The current value of the asset without the "dividends" that will be missed is:

$$S^x = S - PV(Div) = 625\ MNOK - \frac{50\ MNOK}{1.1} = 579.5\ MNOK$$

- The present value of the cost to install the wind park in one year is:

$$PV(K) = \frac{600\ MNOK}{1.04} = 577\ MNOK$$

- The current value of the call option to invest in and start building the wind park is

$$C = S^x N(d_1) - PV(K)N(d_2) = 126.7\ MNOK$$

Investment as a Call Option

- The value today from waiting to invest in the wind park next year (and only start building it if it is profitable to do so) is 126.7 MNOK.
- This exceeds the NPV of 25 MNOK from investing in the wind park today.
- Thus, you are better off waiting to invest, and the value of the contract is 126.7 MNOK.

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Investment as a Call Option

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Investment as a Call Option

- What is the advantage of waiting in this case?
  - If you wait, you will learn more about the likely success of the business.
  - Because the investment in the wind park is not yet committed, you can cancel your plans if the electricity prices should go down or other factors affect the project's prospective success in a negative way.
  - By investing today, you give up this option to "walk away."
- Whether it is optimal to invest today or in one year will depend on the *magnitude of any lost profits* from the first year, compared to the *benefit of preserving your right* to change your decision.

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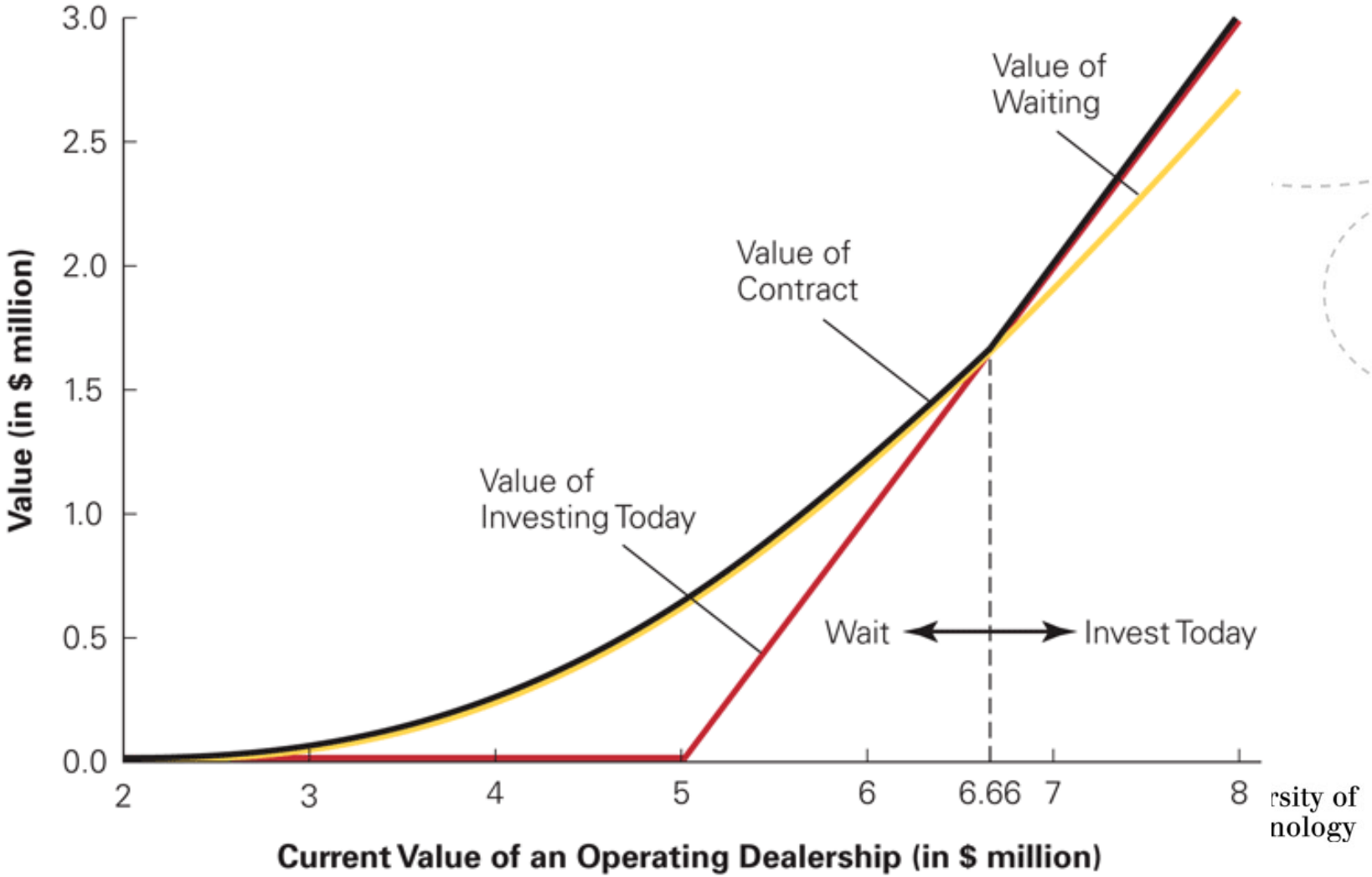
The Decision to Invest in the Wind Park



Factors Affecting the Timing of Investment

- When you have the option of deciding when to invest, it is usually optimal to invest only when the NPV is substantially greater than zero.
  - You should invest today only if the NPV of investing today exceeds the value of the option of waiting.
  - Given the option to wait, an investment that currently has a negative NPV can have a positive one.

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Factors Affecting the Timing of Investment

- Volatility
  - The option to wait is most valuable when there is a great deal of uncertainty.
- Dividends
  - Absent dividends, it is not optimal to exercise a call option early.
  - In the real option context, it is always better to wait unless there is a cost to doing so.
  - The greater the cost, the less attractive the option to delay becomes.

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Growth and Abandonment Options

- Growth Option
  - A real option to invest in the future
  - Future growth opportunities can be thought of as a collection of real call options on potential projects.
  - This can explain why young firms tend to have higher returns than established firms.
- Abandonment Option
  - The option to disinvest
- Because these options have value, they contribute to the value of any firm that has future possible investment opportunities.

Bigger picture: Investment valuation under uncertainty

1. Deterministic DCF

Does not account for uncertainty

2. Sensitivity analysis

Consider multiple futures

3. Monte Carlo simulation

Model the risks

4. Real options approach

Account for flexibility and value of information

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Discounted cash flow

- DCF is a static view on an investment opportunity

1. Deterministic DCF → 2. Sensitivity analysis → 3. Monte Carlo simulation → 4. Real options approach

Does not account for uncertainty

	0	1	2	3	4
Price	$50	$50	$50	$50	$50
Output	0	5	4	3	2
CAPEX	$100	-	-	-	-
OPEX (per unit)	-	$40	$40	$40	$40
Cash flow (CF)	-$100	$50	$40	$30	$20

"Base" case, price = $50

r = 10%

NPV = $13.6

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Sensitivity analysis

1. Deterministic DCF → 2. Sensitivity analysis → 3. Monte Carlo simulation → 4. Real options approach

Consider multiple futures

	0	1	2	3	4
Price	$50	$70	$70	$70	$70
Output	0	5	4	3	2
CAPEX	$100	-	-	-	-
OPEX (per unit)	-	$40	$40	$40	$40
Cash flow (CF)	-$100	$150	$120	$90	$60

"High" case, price = $70 **NPV = $244**

	0	1	2	3	4
Price	$50	$30	$30	$30	$30
Output	0	5	4	3	2
CAPEX	$100	-	-	-	-
OPEX (per unit)	-	$40	$40	$40	$40
Cash flow (CF)	-$100	-$50	-$40	-$30	-$20

"Low" case, price = $30 **NPV = -$215**

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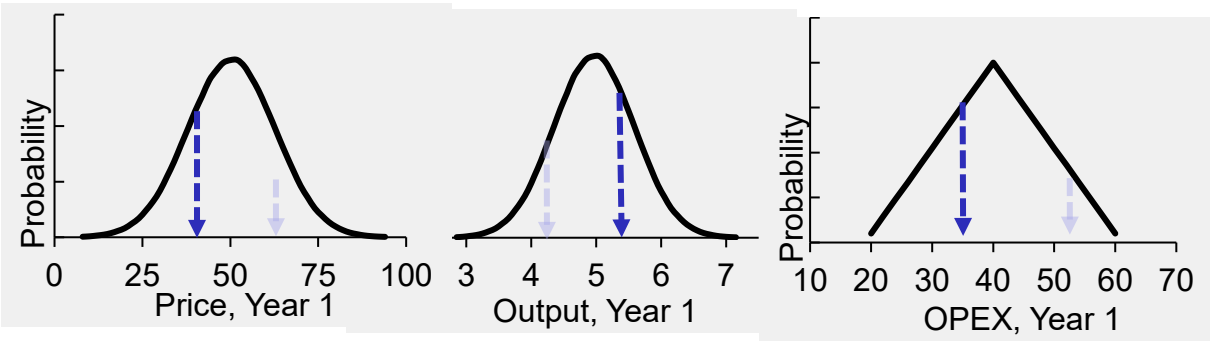
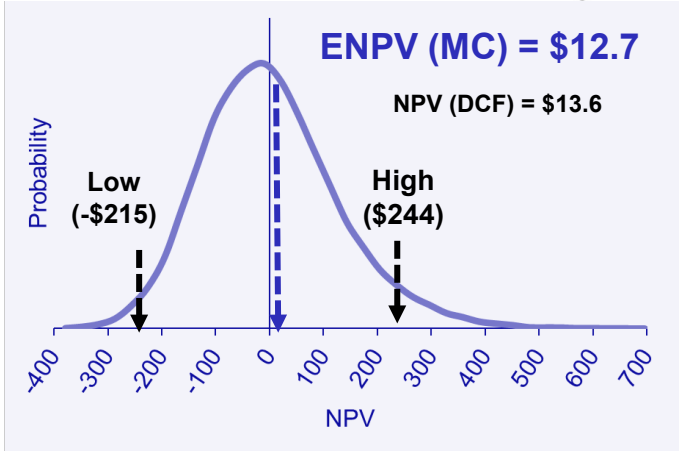
Monte Carlo simulation

1. Deterministic DCF → 2. Sensitivity analysis → 3. Monte Carlo simulation → 4. Real options approach

Model the risks





	Price, Y1	Output, Y1	Costs, Y1	Cash flow, Y1
Sample 1	$45	5.2	$35	$52
Sample 2	$70	4.1	$55	$61.5

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Real options approach – model and exploit the flexibility

1. Deterministic DCF → 2. Sensitivity analysis → 3. Monte Carlo simulation → 4. Real options approach

Account for flexibility and value of information

Price increases

Price drops

	0	1	2	3	4
Price	$50	$100	$50	$30	$10
Output	0	~~5~~ 6	4	3	~~1~~ 0
CAPEX	$100	-	-	-	$5
OPEX (per unit)	-	$42	$40	$40	$40
Cash flow (CF)	-$100	$348	$40	-$30	$-5

Increase output to benefit from higher prices

Stop production to prevent losses

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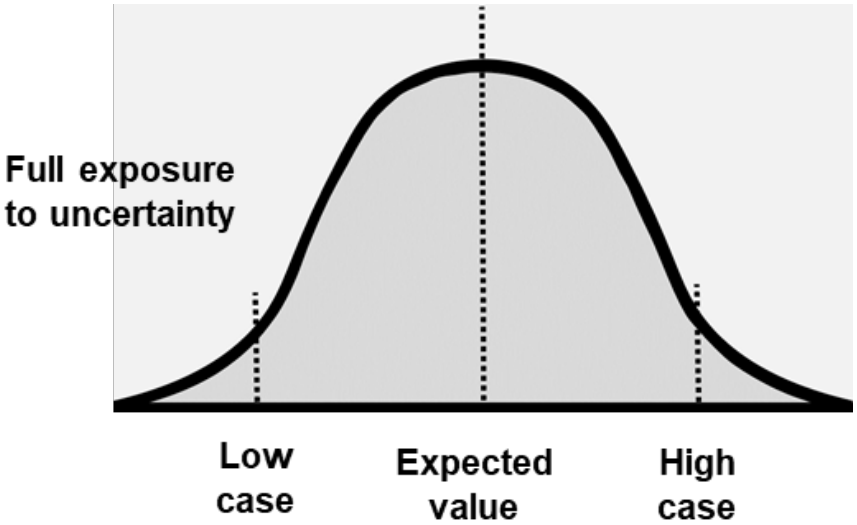
ROA vs. DCF Monte Carlo

DCF and Monte Carlo simulation



Real options approach

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Key Insights from Real Options

- Out-of-the-money real options have value
  - Even if an investment has a negative NPV, if there is a chance it could be positive in the future, the opportunity is worth something today.
- In-the-money real options need not be exercised immediately
  - The option to delay may be worth more than the NPV of undertaking the investment immediately.
- Waiting is valuable
  - By waiting for uncertainty to resolve you can make better decisions.
- Delay investment expenses as much as possible
  - Committing capital before it is absolutely necessary gives up the option to make a better decision once uncertainty is resolved.
- Create value by exploiting real options
  - The firm must continually re-evaluate its investment opportunities, including the options to delay or abandon projects, as well as to create or grow them.